



Air University
Mid Examination : Fall 2025

Student ID: _____

Paper Type: Subjective

Note: To be solved on Answer book only

Time Allowed: 2 Hours

Subject: Artificial Intelligence

Course Code: CS3304

Class: Fall-23

Section(s): A

Special Instructions:

Total Time Allowed: 02 Hrs

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FM Signature:

INSTRUCTIONS

- **Rough work or writing on question paper will be considered as use of unfair means.**
- **Tables/calculators allowed/not allowed.**

Q1. (CLO-01 & PLO-01: Explain artificial intelligence, terminologies, its characteristics and its application areas.) Marks (20)

- a) **Define** Artificial Intelligence (AI) and the purpose of the Turing Test.
- b) **Define** an Agent in the context of AI.
- c) **Define** Simple Reflex Agent and Goal-Based Agent with a diagram.
- d) **Name** the four components of the **PEAS Model** and explain the function of each component.

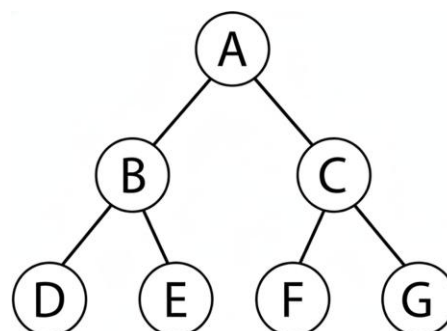
Q2. (CLO-02 & PLO-02: Solving search problems using various uninformed and informed search algorithms. Apply local search algorithms like Genetic Algorithm (GA) on optimization problems and perform Minimax search on games.) Marks (30)

- a) **Explain** Uninformed Search. Also define Depth-First Search (DFS) and specify the data structure used by DFS to manage its frontier.

Explain the following properties of DFS

- ◆ Completeness
- ◆ Optimality (considering both positive and negative step costs)
- ◆ Time Complexity

Apply DFS on the tree given below. Starting from the initial node A, trace the execution of DFS until the goal node G is found. List the exact sequence of expanded nodes and the final solution path returned by DFS.



- b) **Explain** the working of Uniform-Cost Search (UCS) and identify the data structure used by Uniform-Cost Search (UCS) to manage its open list (node list).

Explain the following properties of Uniform-Cost Search (UCS):

- ◆ Completeness
- ◆ Optimality (considering both positive and negative step costs)
- ◆ Time Complexity

Apply UCS on the tree given below. Starting from the initial node S, trace the execution of UCS until the goal node G is found. List the final solution path returned by UCS.

